CVPR 2024, Highlight Paper

Adapting Visual-Language Models for Generalizable Anomaly Detection in Medical Images

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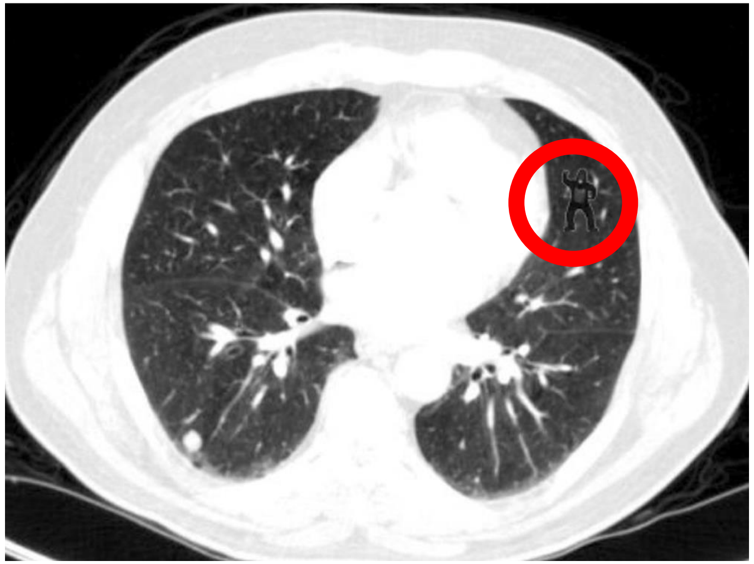
Challenge Task

Anomaly Detection in Medical Images



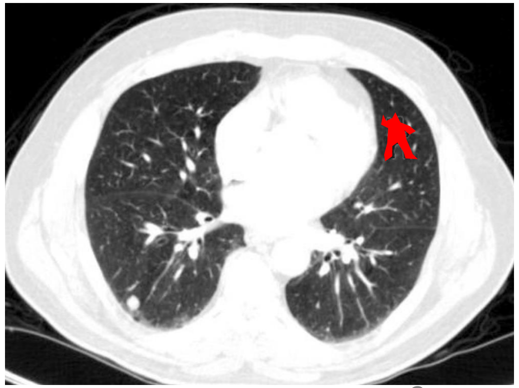
“ Will you be able to spot
a Gorilla in a CT scan?? ”

50% of trained radiologists did not notice a gorilla image

Gives a standardized dataset and benchmark for anomaly detection

1. **Anomaly Classification (AC)**
   In- / Out- of distribution

2. **Anomaly Segmentation (AS)**
   Pixel(Voxel)-level analysis



Challenge Task

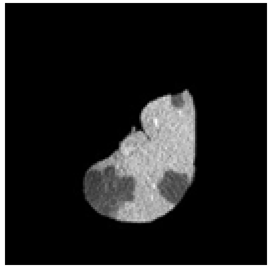
Anomaly Detection in Medical Images

“ Identifying whether a test input is drawn far from the training distribution (in-distribution) or not ”

Training Dataset

Normal Images
: Most of the training dataset consists of normal images

\+



Abnormal Images
: They are included in the training dataset in **very small numbers or none at all**

Compared to the natural dataset,
the total number of images included in the training dataset in the medical domain itself is very small

Challenge Task

Anomaly Detection in Medical Images

1

Full-normal shot

Training Dataset에 포함된 모든 Normal Image만으로 학습시킨 후,

실제 Test에서는 Normal 및 다양한 Abnormal에 대한 판별을 진행하게 된다.

2

Few-normal shot

소수의 Normal Image들만이 Training dataset에 포함되어 Abnormal Image들은 학습 시에 활용되지 않는다.

정상 데이터만 가지고 있으면서, 학습 데이터 수가 부족할 때 활용된다.

3

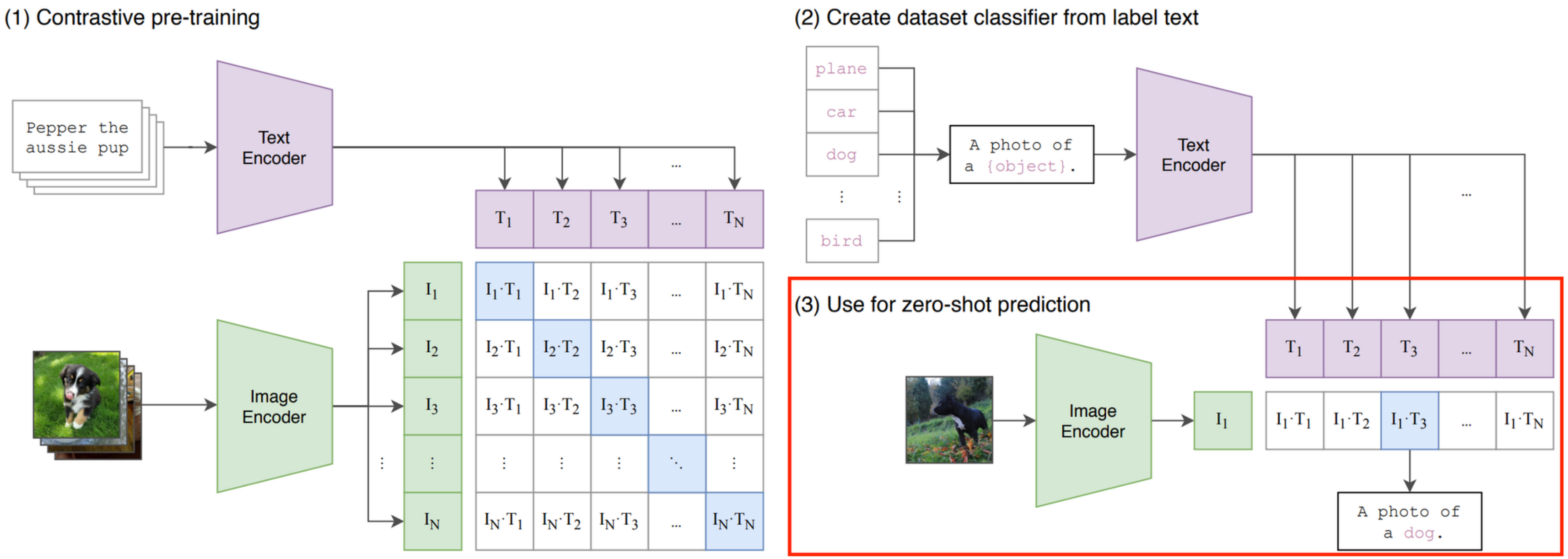
Few-shot

Training Dataset에는 아주 적은 수의 Normal Image와 아주 적은 수의 Abnormal Image들이 포함되어 있다.

Normal과 Abnormal 이미지를 모두 가지고 있으나, 그 수가 부족할 때 사용할 수 있다.

Adaptation in VLMs for Medical Anomaly Detection

CLIP: Pretrained Visual-Language Model



- ✓ generalizability와 robustness이 뛰어나며, language-driven zero-shot inference 가능
- ✓ language-guided detection and segmentation을 위해 pre-trained CLIP models 사용
- ✓ **본 연구는 pretrained VLM을 의료 이미지의 Anomaly Detection(AD)에 확장 시켜 적용함**

Introduction

Adaptation in VLMs for Medical Anomaly Detection

Natural Image Domain / Medical Image Domain

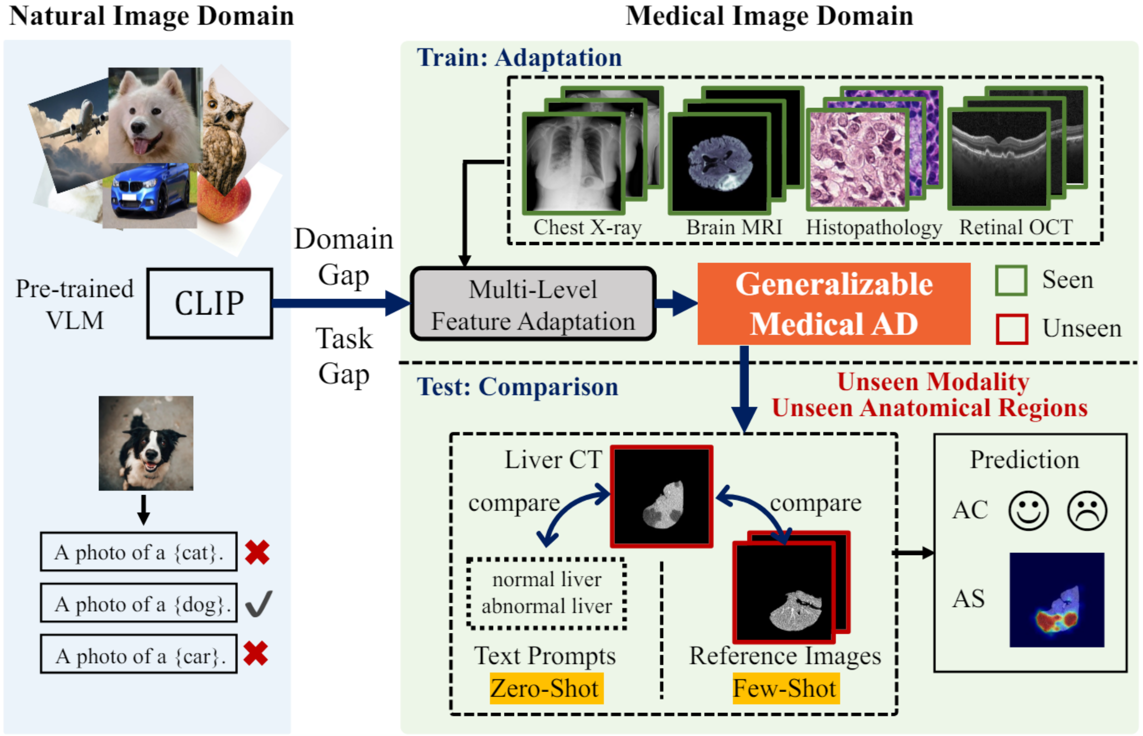


Figure 1. The overview of adaptation in pre-trained visual-language models for zero-/few-shot medical anomaly classification (AC) and anomaly segmentation (AS).

- ✓ CLIP을 포함한 대부분의 VLM은 대용량의 Natural image들로 pretrain되었기에, Domain Gap이나 Task Gap이 큰 분야에 대해서는 좋은 성능을 보이지 못했음
- ✓ 그 중에서도 Medical Image 분야의 Anomaly Detection Task의 경우 Task 및 Domain에서 Natural Image와 많은 차이가 존재해 제대로 된 성능을 내지 못했음
- ✓ 본 논문에서는 **CLIP을 의료영상 분야의 Anomaly Detection (AD)에 활용할 수 있는 일반화된 모델을 개발**하여 Medical Image의 다양한 Modality에서 높은 성능을 낼 수 있도록 함

Method

Multi-level Adaptation

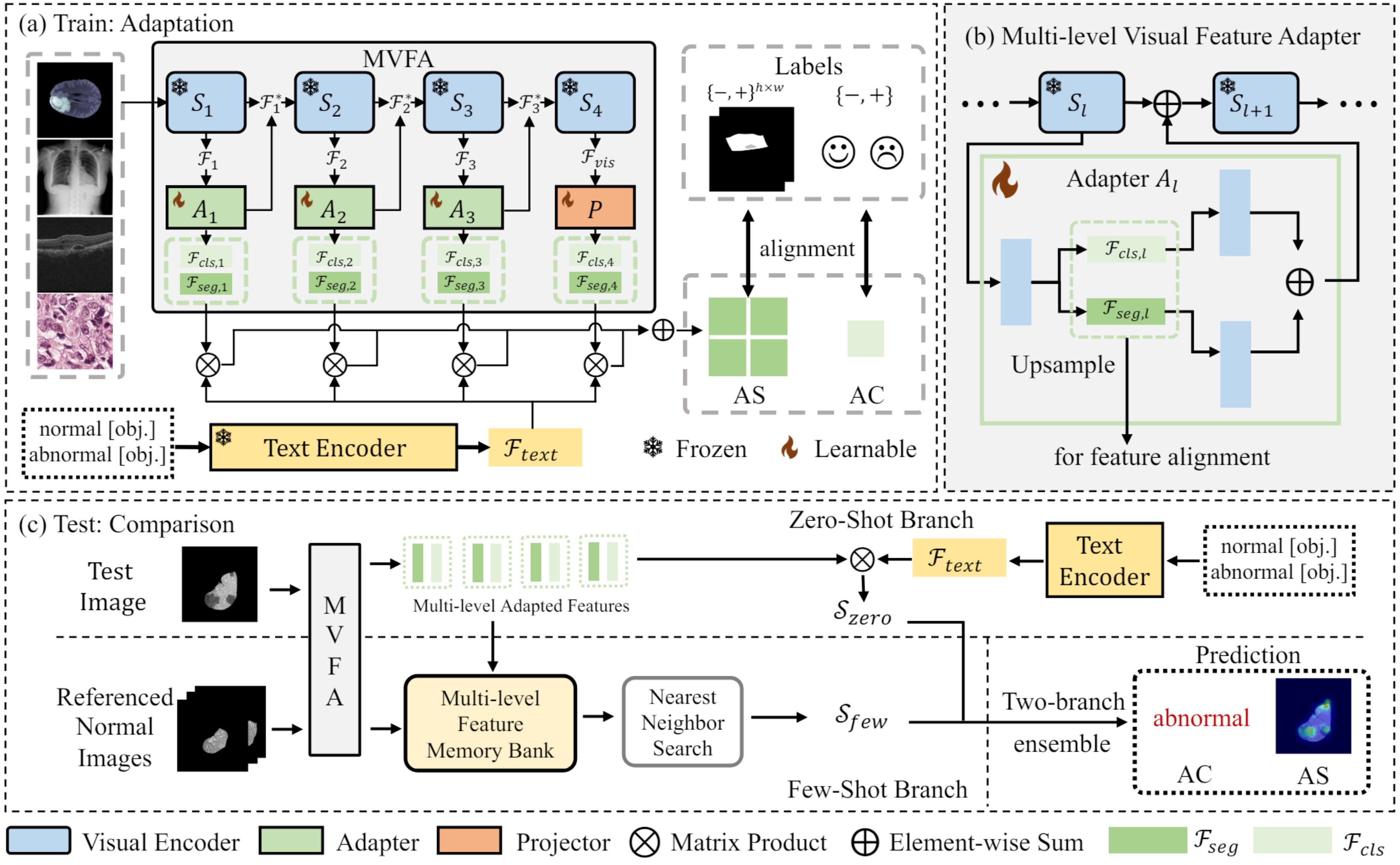


Figure 2. The architecture of multi-level adaptation and comparison framework for zero-/few-shot medical anomaly detection.

Method

Multi-level Adaptation

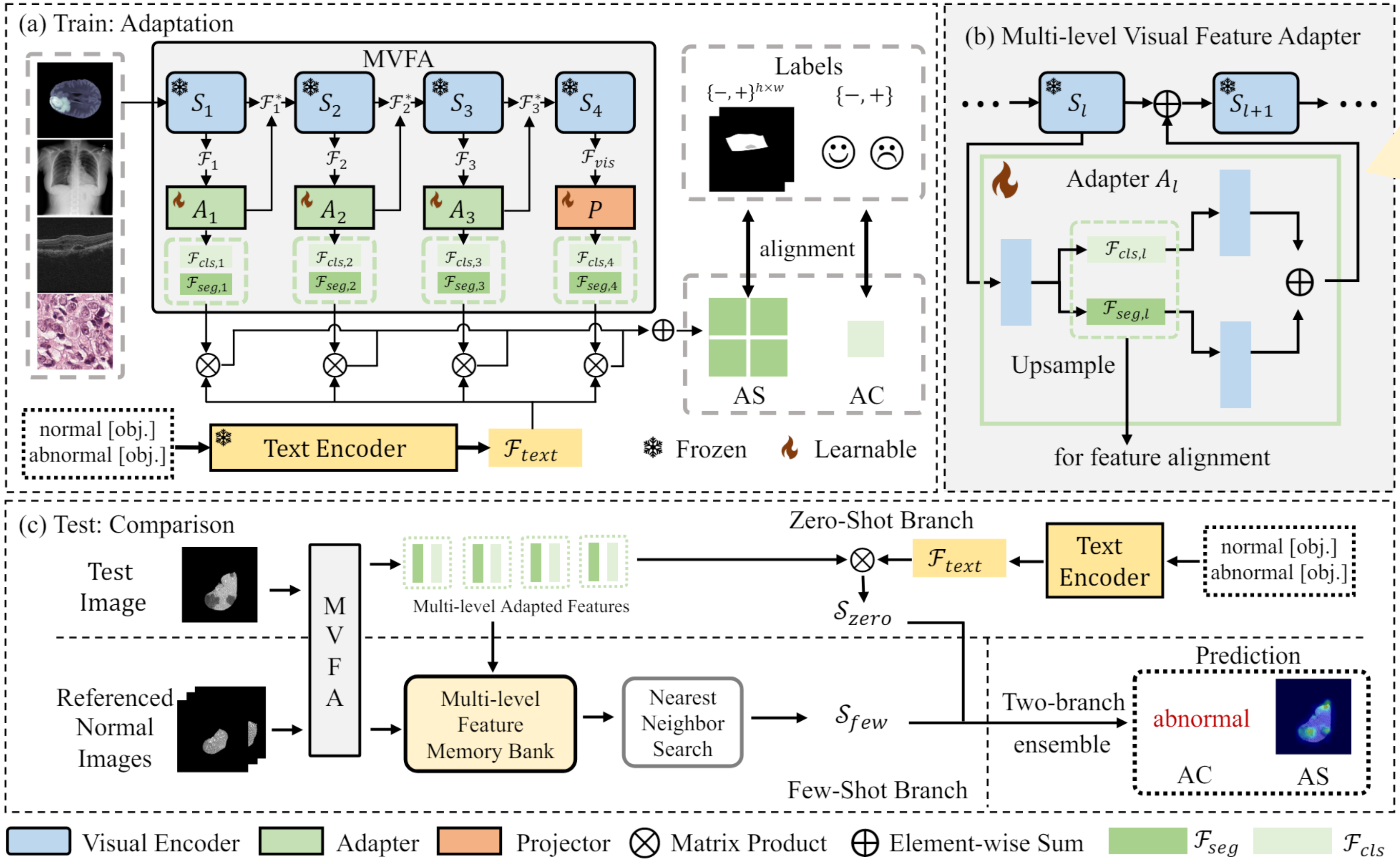


Figure 2. The architecture of multi-level adaptation and comparison framework for zero-/few-shot medical anomaly detection.

Train: Multi-level Feature Adaptation

- ✓ Multi-level Visual Feature Adapter (MVFA)
- ✓ Language Feature Formatting
- ✓ Visual-Language Feature Alignment

Method

Multi-level Adaptation

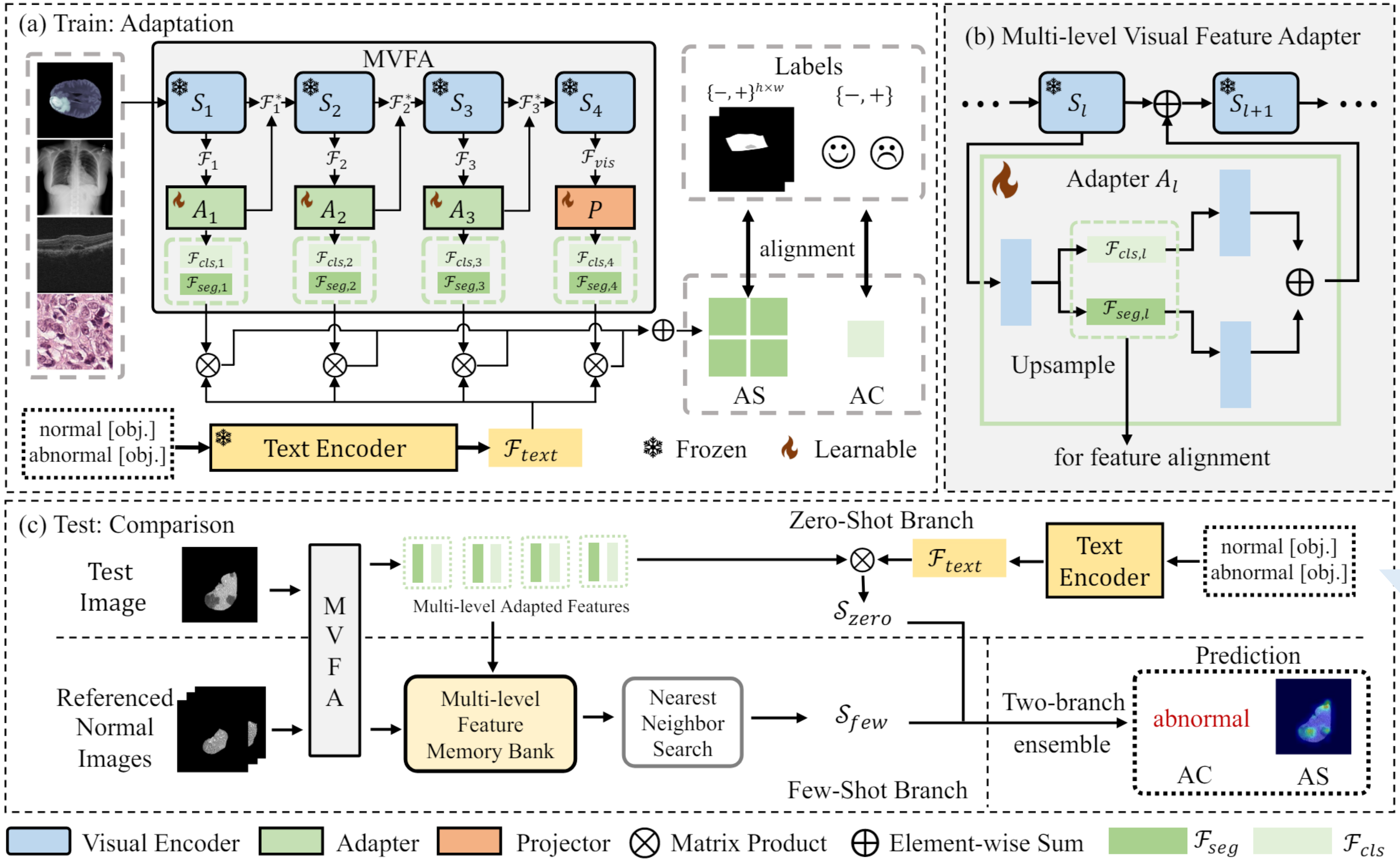


Figure 2. The architecture of multi-level adaptation and comparison framework for zero-/few-shot medical anomaly detection.

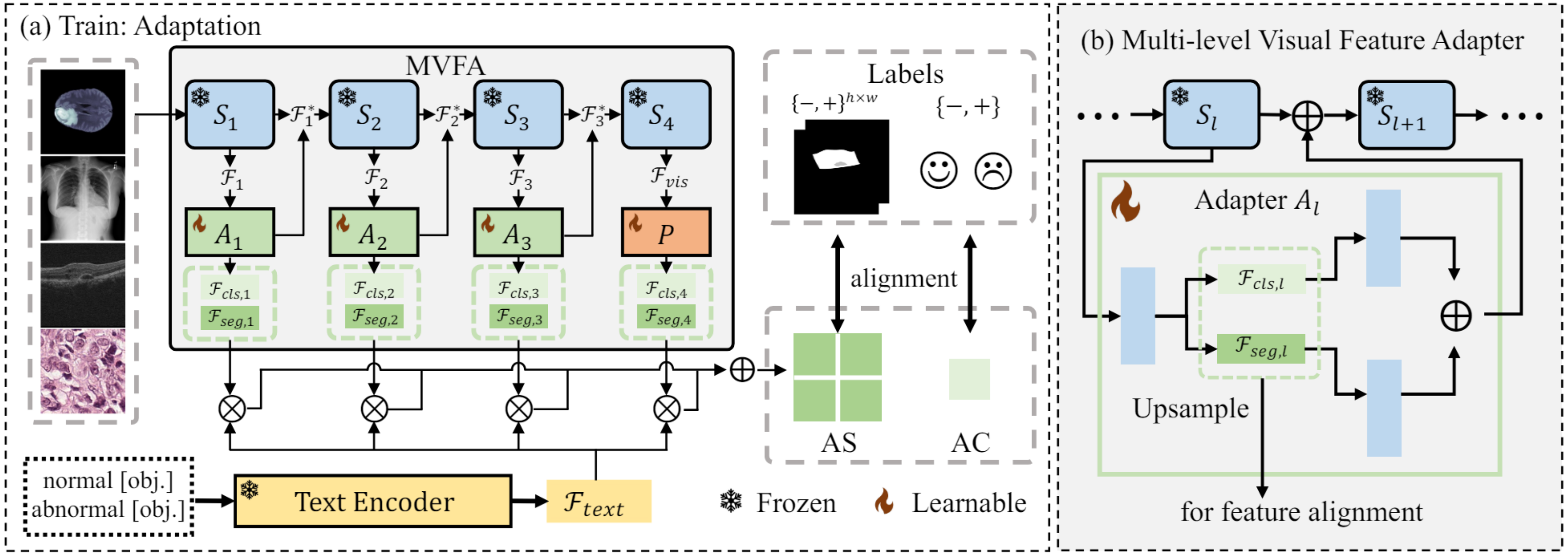
Test: Multi-level Feature Comparison

- ✓ Zero-shot Branch
- ✓ Few-shot Branch

Q & A

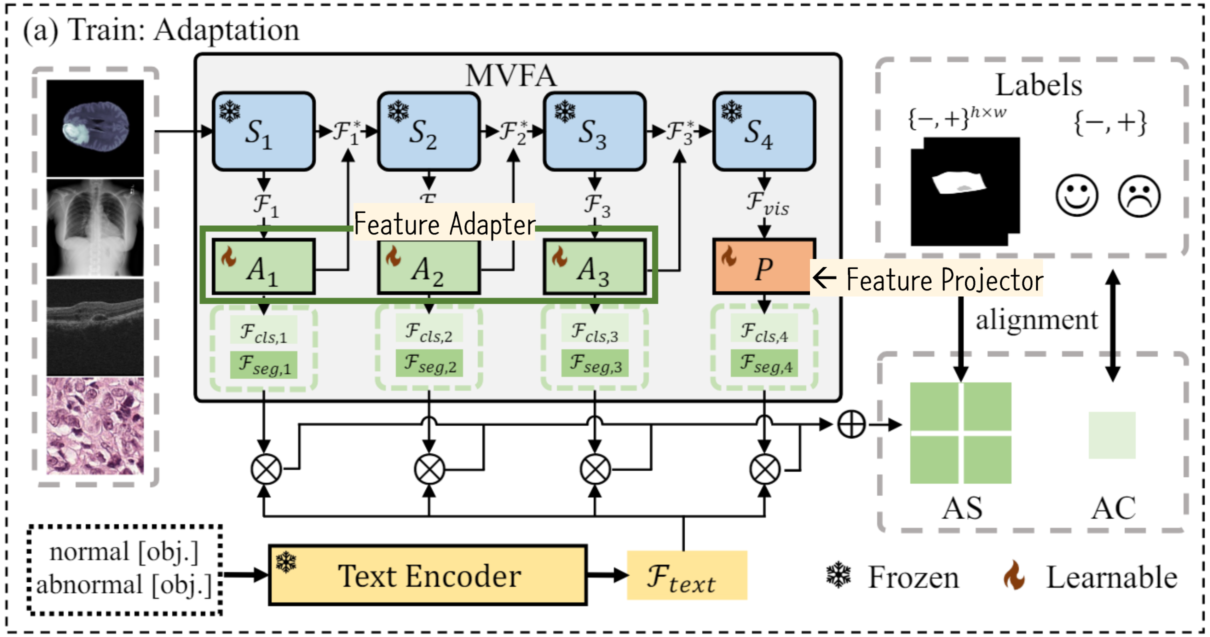
Method

Train: Multi-level Visual Feature Adapter (MVFA)



- ✓ 의료 영상의 AD를 위해 사전 학습된 Visual-Language Model(CLIP)을 활용하는 Multi-level Visual Feature Adaptation 프레임워크의 핵심 구성 요소
- ✓ 의료 영상에서 거대한 모델을 활용하는 경우, 거대한 수의 파라미터에 비해 학습 데이터의 수가 적기 때문에 Overfitting이 발생하는 경우가 많았었음
- ✓ 따라서 MVFA는 Overfitting을 방지하고 Training Sample의 수가 제한된 상황에서 높은 성능을 유지하기 위해 CLIP 모델의 residual adapter에 **적은 수의 학습 가능한 Learnable Bottleneck Linear Layers을 추가하여 Multi-level Feature를 활용**할 수 있도록 함
- ✓ 이 방법은 기존의 CLIP Backbone을 변경하지 않고도 높은 성능을 내는 데에 기여함

Train: Multi-level Visual Feature Adapter (MVFA)



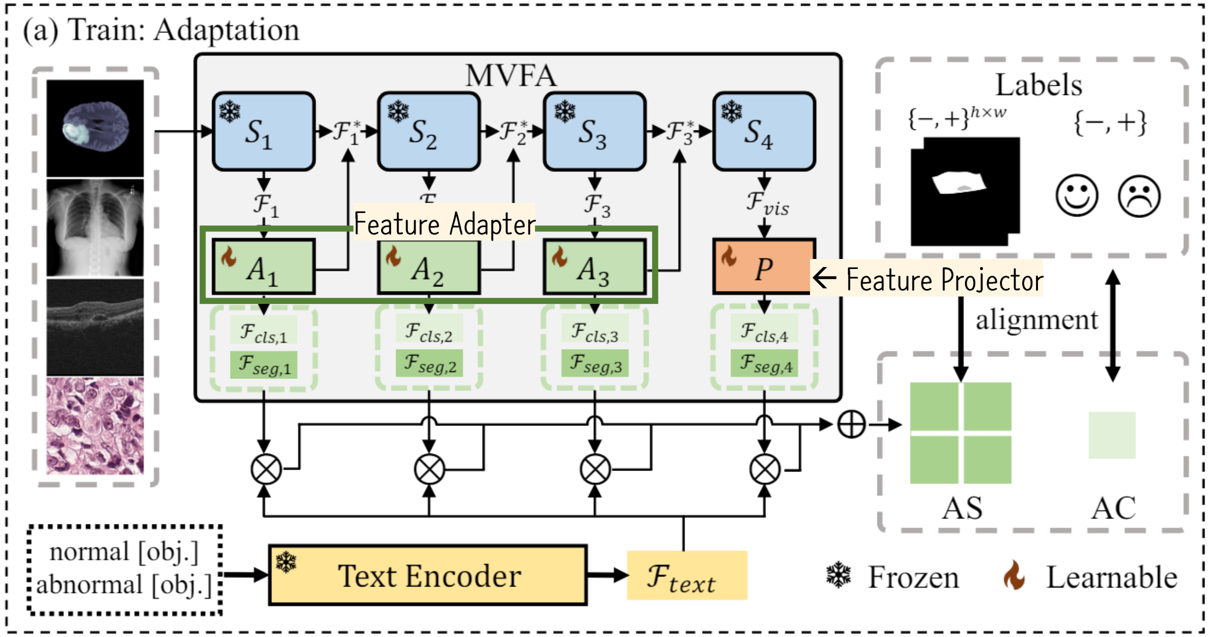
- S_1 to S_4: CLIP의 4개의 sequential stages
- $F_l, l \in 1,2,3$: S_l를 거친 middle stage features
- $A_l, l \in 1,2,3$: learnable feature adapter
- P: feature projector
- $feature\ adaptation$: 3 adaptors (A_l) + 1 projector (P)

feature adaptor A_l은 middle stage feature F_l에 최소 두 개 이상의 trainable linear transformation layer $W_{l,1}$, $W_{l,2}$를 통과시킴

$$A_l(\mathcal{F}_l) = ReLU(\mathcal{F}_l^T W_{l,1})W_{l,2}, \text{where } l \in \{1,2,3\}. \quad (1)$$

Method

Train: Multi-level Visual Feature Adapter (MVFA)



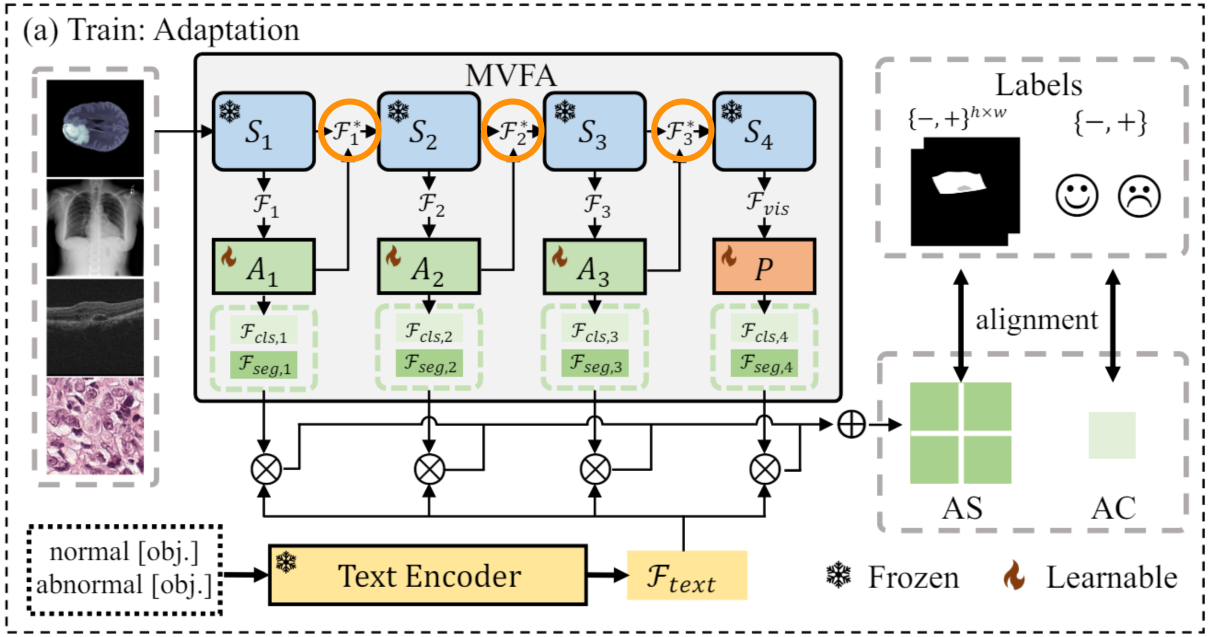

```python
# Residual CLIP Adapter
class ClipAdapter(nn.Module):
    def __init__(self, c_in, bottleneck=768):
        super(ClipAdapter, self).__init__()
        self.fc1 = nn.Sequential(
            nn.Linear(c_in, bottleneck, bias=False),
            nn.LeakyReLU(inplace=False)
        )
        self.fc2 = nn.Sequential(
            nn.Linear(bottleneck, c_in, bias=False),
            nn.LeakyReLU(inplace=False)
        )

    def forward(self, x):
        x = self.fc1(x)
        y = self.fc2(x)
        return x, y
```

feature adaptor A_l은 middle stage feature F_l에 최소 두 개 이상의 trainable linear transformation layer $W_{l,1}$, $W_{l,2}$를 통과시킴

$$A_l(\mathcal{F}_l) = ReLU(\mathcal{F}_l^T W_{l,1})W_{l,2}, \text{where } l \in \{1, 2, 3\}. \quad (1)$$

Method

Train: Multi-level Visual Feature Adapter (MVFA)



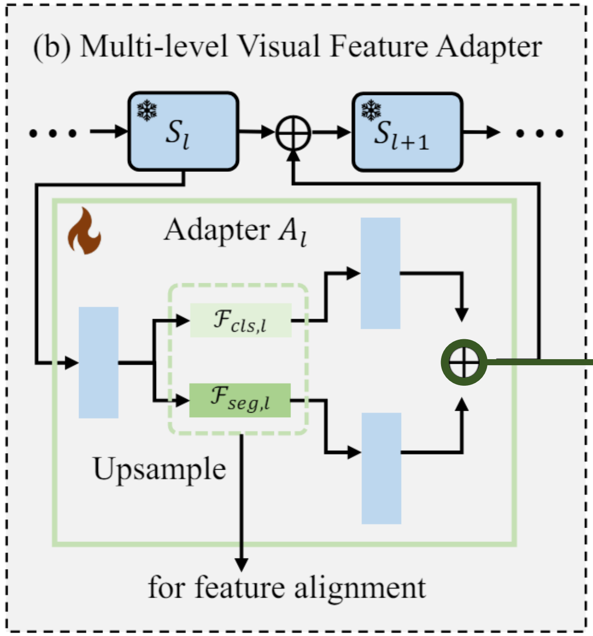
- S_1 to S_4: CLIP의 4개의 sequential stages
- $F_l, l \in 1,2,3$: S_l를 거친 middle stage features
- $A_l, l \in 1,2,3$: learnable feature adapter
- P: feature projector
- *feature adaptation*: 3 adaptors (A_l) + 1 projector (P)

middle stage feature들을 합칠 때는 Residual Ratio값인 gamma를 활용하여 Weighted Sum을 수행함 (by default, gamma = 0.1)

$$\mathcal{F}_l^* = \gamma A_l(\mathcal{F}_l)^T + (1-\gamma)\mathcal{F}_l, \text{where } l \in \{1,2,3\}, \quad (2)$$

Method

Train: Multi-level Visual Feature Adapter (MVFA)



AC를 위한 **global feature**와 AS를 위한 **local feature**를 동시에 다루기 위해서 feature adaptor를 parallel한 dual-architecture 형태로 적용 시킴

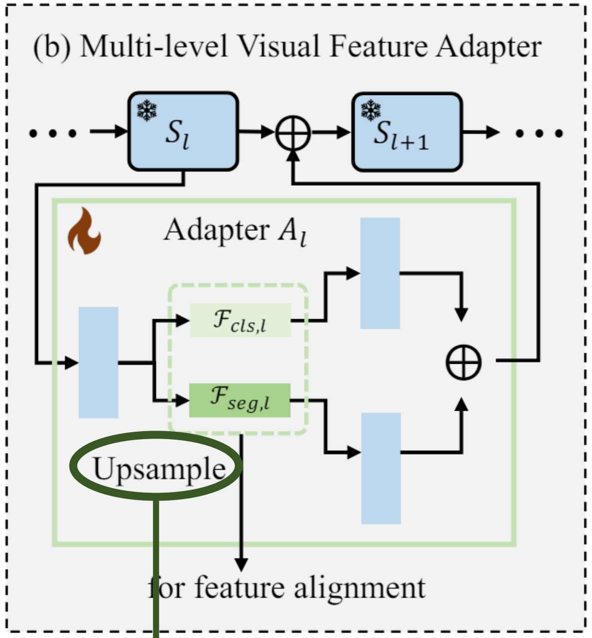
```
self.seg_adapters = nn.ModuleList( [ClipAdapter(1024, bottleneck=768) for i in range(len(features))] )
self.det_adapters = nn.ModuleList( [ClipAdapter(1024, bottleneck=768) for i in range(len(features))] )
```

- global feature (for AC): $F_{cls,4} = F_{vis}^T W_{cls}$
- local feature (for AS): $F_{seg,4} = F_{vis}^T W_{seg}$

```
for i in range(24):
    if i + 1 == 12:
        x, attn = self.image_encoder.transformer.resblocks[i](x, attn_mask=None)
        attn_out.append(attn)
    else:
        x, attn_map = self.image_encoder.transformer.resblocks[i](x, attn_mask=None)
    if (i + 1) in self.features:
        seg_adapt_med, seg_adapt_out = self.seg_adapters[self.features.index(i+1)](x)
        det_adapt_med, det_adapt_out = self.det_adapters[self.features.index(i+1)](x)

        x = 0.8 * x + 0.1 * seg_adapt_out + 0.1 * det_adapt_out

        seg_patch_tokens.append(seg_adapt_med)
        det_patch_tokens.append(det_adapt_med)
```

Method

Train: Multi-level Visual Feature Adapter (MVFA)



AC를 위한 **global feature**와 AS를 위한 **local feature**를 동시에 다루기 위해서 feature adaptor를 parallel한 dual-architecture 형태로 적용 시킴

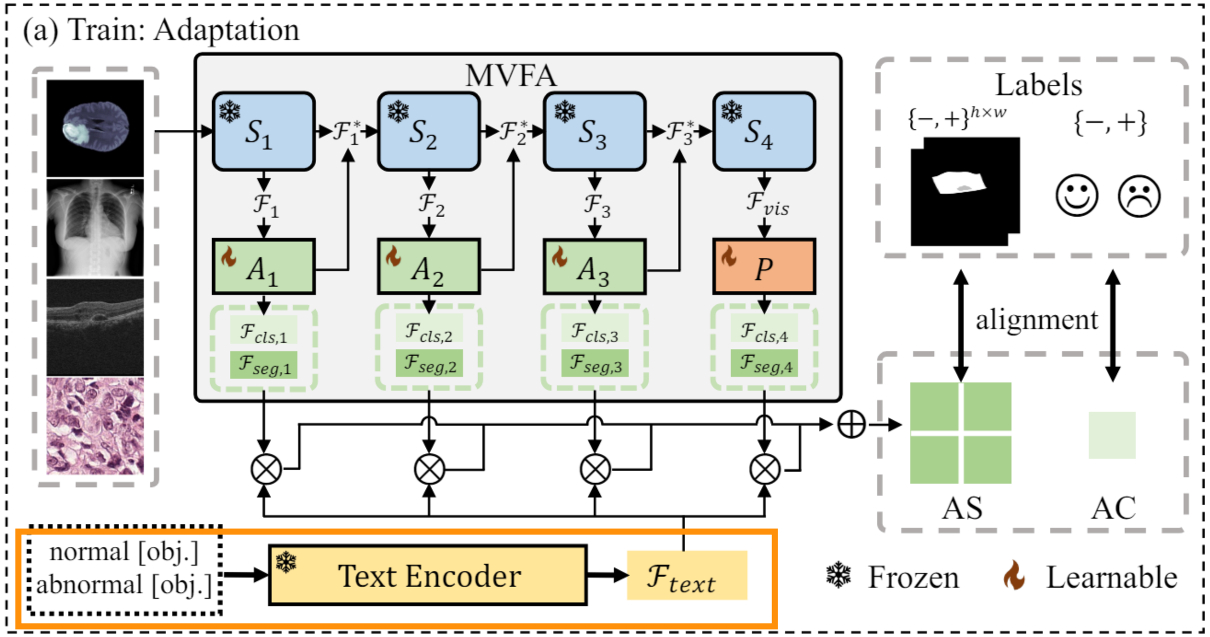
- global feature (for AC): $F_{cls,4} = F_{vis}^T W_{cls}$
- local feature (for AS): $F_{seg,4} = F_{vis}^T W_{seg}$

(Upsampling 부분은 Segmentation Task의 Loss 계산 시에 활용됨)

```
# few-shot, seg head
anomaly_maps_few_shot = []
for idx, p in enumerate(seg_patch_tokens):
    cos = cos_sim(seg_mem_features[idx], p)
    height = int(np.sqrt(cos.shape[1]))
    anomaly_map_few_shot = torch.min((1 - cos), 0)[0].reshape(1, 1, height, height)
    anomaly_map_few_shot = F.interpolate(torch.tensor(anomaly_map_few_shot),
                                         size=args.img_size, mode='bilinear', align_corners=True)
    anomaly_maps_few_shot.append(anomaly_map_few_shot[0].cpu().numpy())
score_map_few = np.sum(anomaly_maps_few_shot, axis=0)
seg_score_map_few.append(score_map_few)
```

Method

Train: Language Feature Formatting



(a) *State*-level (-:normal, +:abnormal)

```
- c := "[o]"
- c := "flawless [o]"
- c := "perfect [o]"
- c := "unblemished [o]"
- c := "[o] without flaw"
- c := "[o] without defect"
- c := "[o] without damage"
+ c := "damaged [o]"
+ c := "[o] with flaw"
+ c := "[o] with defect"
+ c := "[o] with damage"
```

(b) *Template*-level

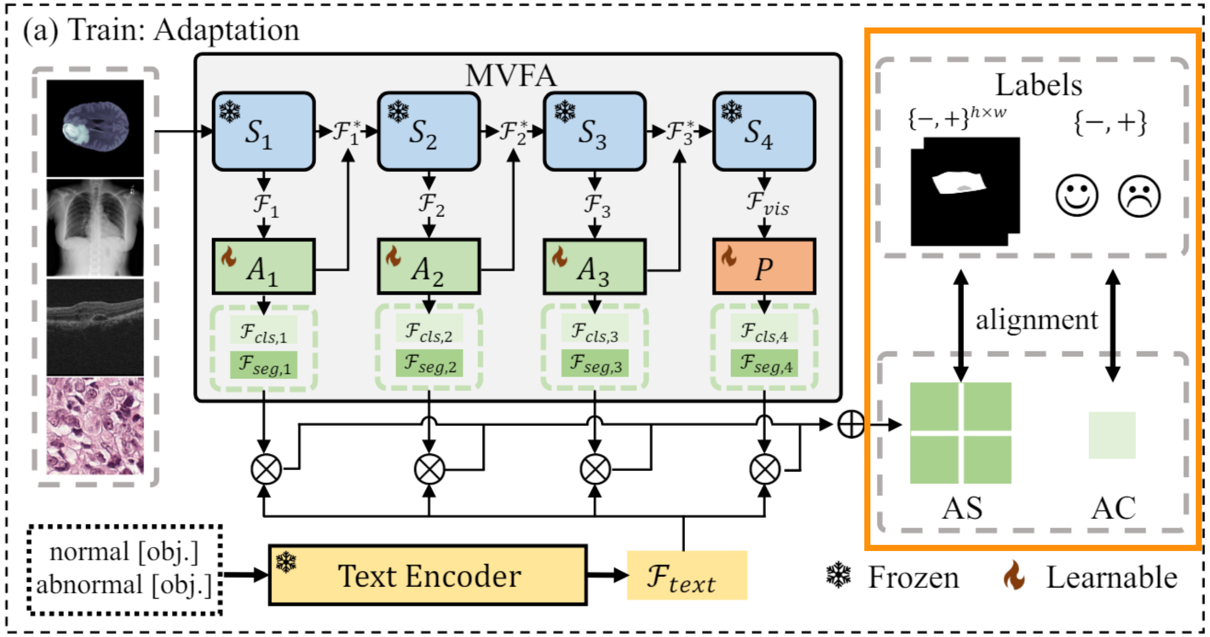
- "a photo of a/the/one [c]."
- "a photo of a/the cool [c]."
- "a photo of a/the small [c]."
- "a photo of a/the large [c]."
- "a bright photo of a/the [c]."
- "a dark photo of a/the [c]."
- "a blurry photo of a/the [c]."
- "a bad photo of a/the [c]."
- "a good photo of a/the [c]."
- "a cropped photo of a/the [c]."
- "a close-up photo of a/the [c]."
- (cont'd) "a photo of my [c]."
- "a low resolution photo of a/the [c]."
- "a black and white photo of a/the [c]."
- "a jpeg corrupted photo of a/the [c]."
- "there is a/the [c] in the scene."
- "this is a/the/one [c] in the scene."

✓ CLIP이 대규모 Visual Language Model이기 때문에 normal과 anomaly에 관련된 text prompt를 활용함

✓ 복잡한 디테일에 관련된 text prompt보다는 명료한 text prompt를 주려고 했고, 예시와 같은 35개의 text prompt template을 활용하여 학습 진행

Method

Train: Visual-Language Feature Alignment



$$\mathcal{L}_l = \lambda_1 Dice(softmax(\mathcal{F}_{seg,l}\mathcal{F}_{text}^T), \mathcal{S}) + \lambda_2 Focal(softmax(\mathcal{F}_{seg,l}\mathcal{F}_{text}^T), \mathcal{S}) + \lambda_3 BCE(\max_{h\times w}(softmax(\mathcal{F}_{cls,l}\mathcal{F}_{text}^T)), c), \quad (3)$$

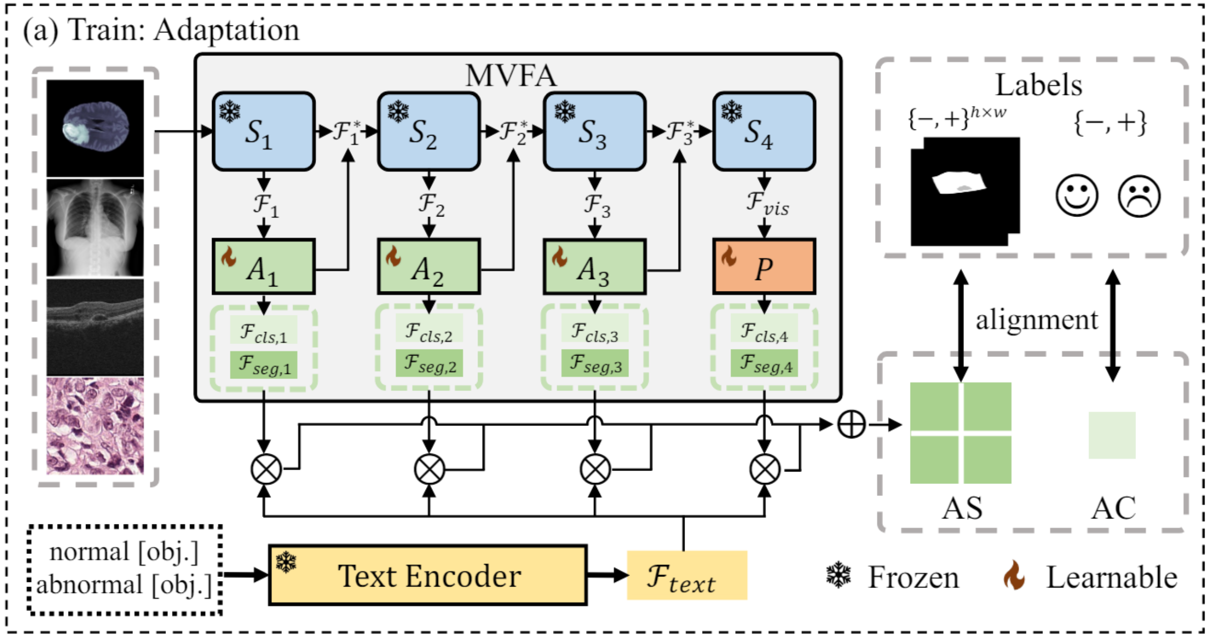
- global feature (for AC): $F_{cls,4} = F_{vis}^T W_{cls}$
- local feature (for AS): $F_{seg,4} = F_{vis}^T W_{seg}$
- $F_{text} \in R^{2\times d}$ where d is the feature dimension

✓ 각 feature level (l∈{1,2,3,4})마다 MVFA와 text feature들 간의 alignment를 맞춰주며 모델을 optimize해 줌

✓ 여기서 λ값은 모두 1.0으로 동일하게 설정하여, 최종 loss인 L_{adapt}는 모든 level에서의 L_l를 합산한 값이 됨

Method

Train: Discussion



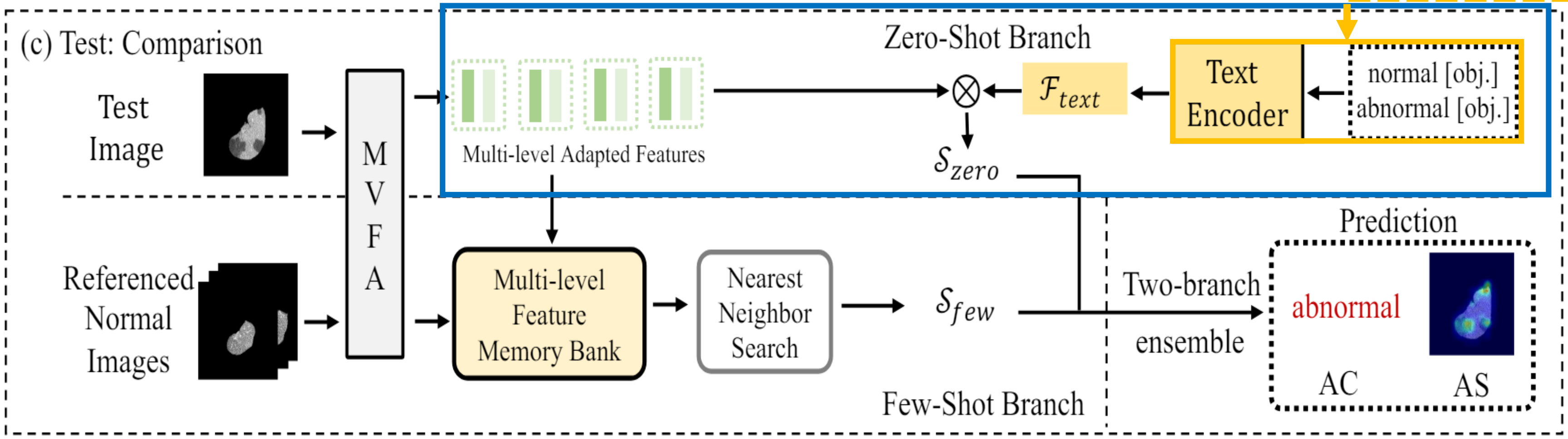
Good at

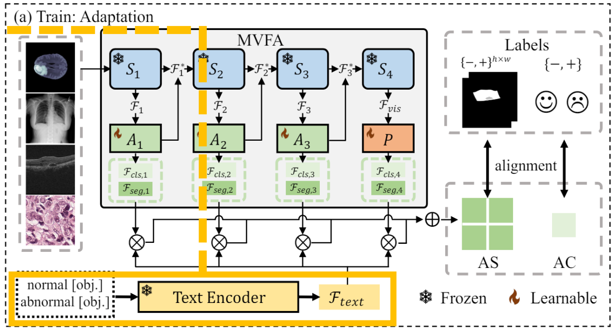
Generalization

- **WinCLIP**
  : pre-trained VLM의 class token만을 활용함
- **APRIL-GAN**
  : projection된 이후의 feature만 alignment을 맞춰 줌
- **MVFA**
  : 여러 stage마다 multi-level adaptation을 도입하여 visual-language alignment를 수행하여 medical anomaly detection에서 robust한 결과를 낼 수 있도록 함

Method

Test: Zero-shot Branch





✓ MVFA를 거쳐 얻은 multi-level adapted feature를 **normal과 abnormal text prompt들에 대한 feature들을 training 시에 저장해 둔 $\mathcal{F}_{text}$와 비교함**

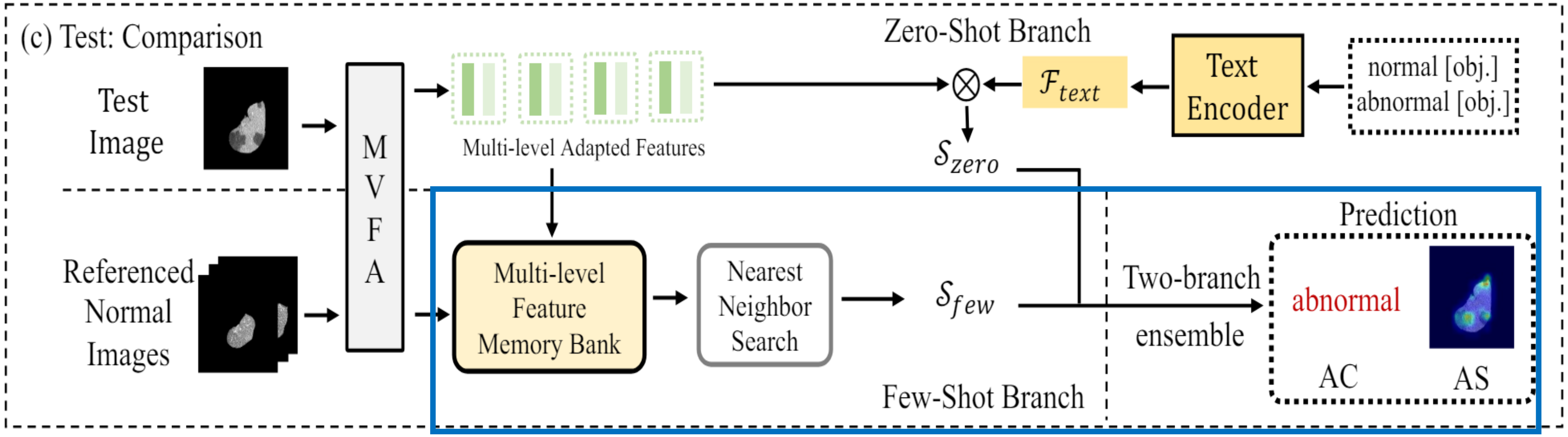
✓ 최종적으로 zero-shot의 AC 결과(c_{zero})와 AS 결과(S_{zero})는 각 level마다 나오는 결과들에 대해서 average softmax를 취하여 얻을 수 있게 된다.

$$c_{\text{zero}} = \frac{1}{4}\sum_{l=1}^{4} \max_{G}(softmax(\mathcal{F}_{cls,l}\mathcal{F}_{text}^{T})),$$
$$S_{\text{zero}} = \frac{1}{4}\sum_{l=1}^{4}\text{BI}(softmax(\mathcal{F}_{seg,l}\mathcal{F}_{text}^{T})). \quad (4)$$

Here, BI$(\cdot)$ reshapes the anomaly map to $\sqrt{G}\times\sqrt{G}$ and restores it to the original input image resolution using bilinear interpolation, with G representing the grid number.

Method

Test: Few-shot Branch



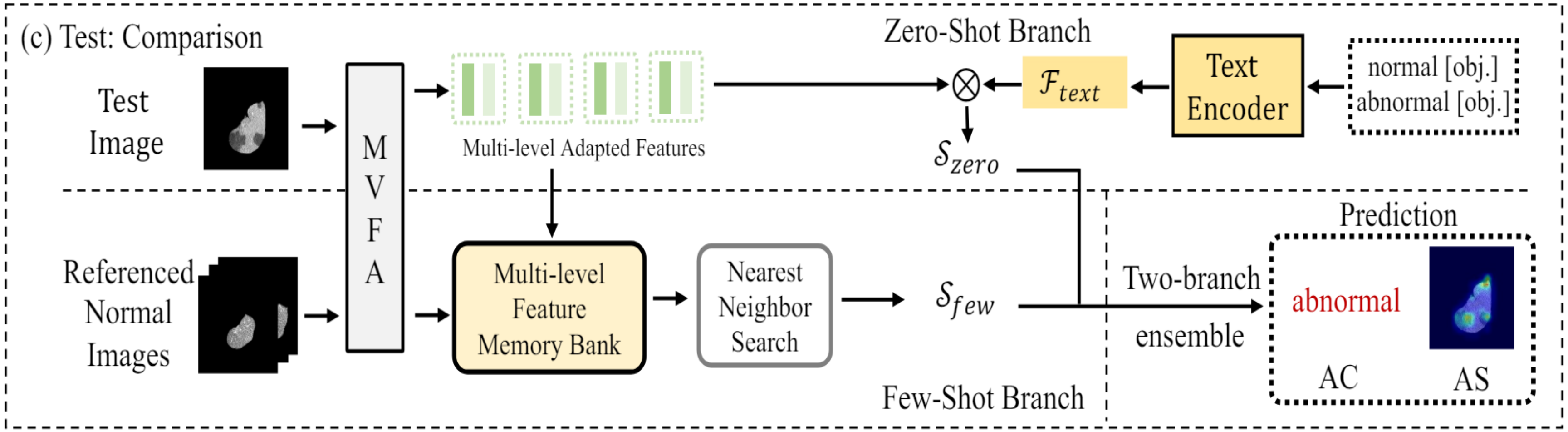
- ✓ Training Dataset D_{med}에 포함되어 있는 labeled normal image들의 multi-level visual feature들은 **Feature Memory Bank G에 포함됨**
- ✓ 따라서 Few-shot AC(c_{few})와 AS(S_{few}) score는 feature memory bank G와 multi-level adapted feature 간의 distance를 계산하여 얻음

$$\begin{aligned} c_{\text{few}} &= \frac{1}{4}\sum_{l=1}^{4} \max_{G}(\min_{m\in\mathcal{G}} Dist(\mathcal{F}_{cls,l}, m)), \\ \mathcal{S}_{\text{few}} &= \frac{1}{4}\sum_{l=1}^{4} \text{BI}(\min_{m\in\mathcal{G}} Dist(\mathcal{F}_{seg,l}, m)). \end{aligned} \quad (5)$$

Here, $Dist(\cdot,\cdot)$ represents the cosine distance, calculated as $1 - \text{cosine}(\cdot,\cdot)$. The final predicted AC and AS results combine the outcomes from both branches:

Method

Test

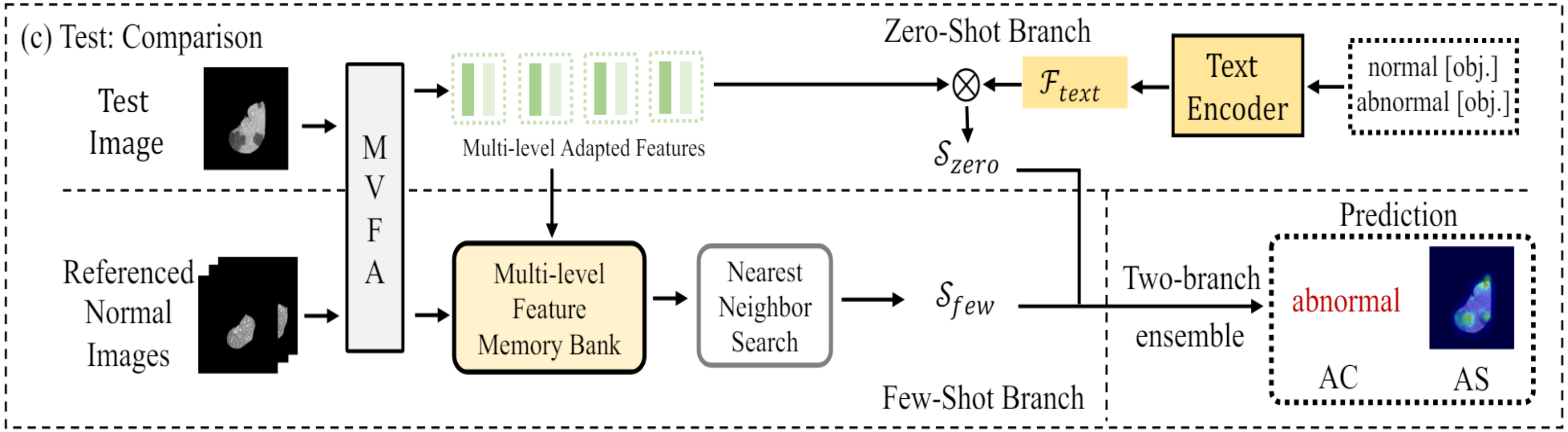


✓ 최종적으로 test 시의 AC(c_{pred})와 AS($\mathcal{S}_{pred}$)는 zero-shot branch와 few-shot branch의 결과를 평균 낸 값이 됨

$$c_{\text{pred}} = \beta_1 c_{\text{zero}} + \beta_2 c_{\text{few}}, \mathcal{S}_{\text{pred}} = \beta_1 \mathcal{S}_{\text{zero}} + \beta_2 \mathcal{S}_{\text{few}}. \quad (6)$$

Here, β_1 and β_2 are weighting factors for the zero-shot and few-shot branches, respectively, set to 0.5 by default.

Method
Test



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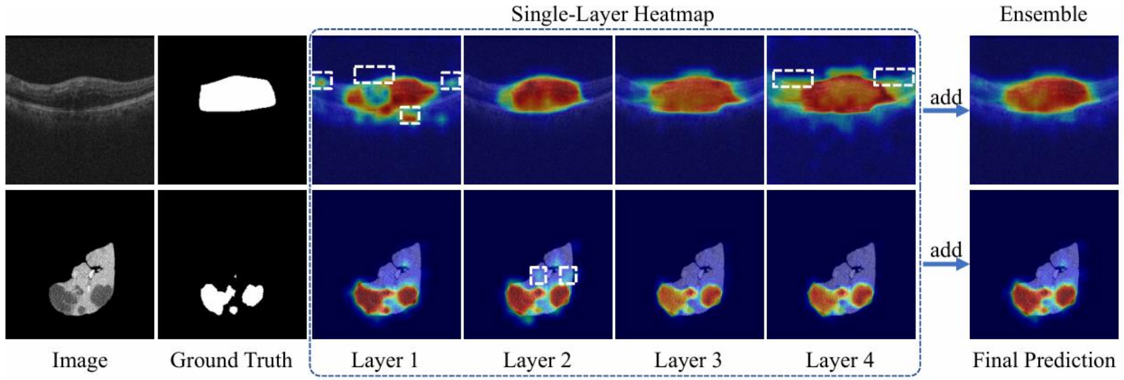
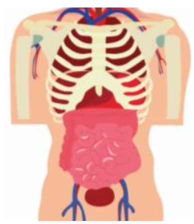


Figure 9. Visualization of anomaly segmentation heatmaps from the four single layers and the multi-layer ensemble results. The white dashed boxes demarcate regions that have been missed or erroneously segmented.

Q & A



Experiments

Experimental Setups

Dataset

- D_{med}: medical training dataset, which consists of annotated samples from the medical field. it is defined as a set of tuples $(x_i, c_i, S_i)_{i=1}^{K}$
  - K: the total number of image samples in the dataset
  - pre-training dataset that is composed of medical data from different modalities and anatomical regions than those in the test samples, which assesses the model's generalization to unseen scenarios.
  - includes a small collection of K annotated images that are of the same modality and anatomical region as those in the test samples, with K typically representing a small numerical value, such as {2, 4, 8, 16} in this study.

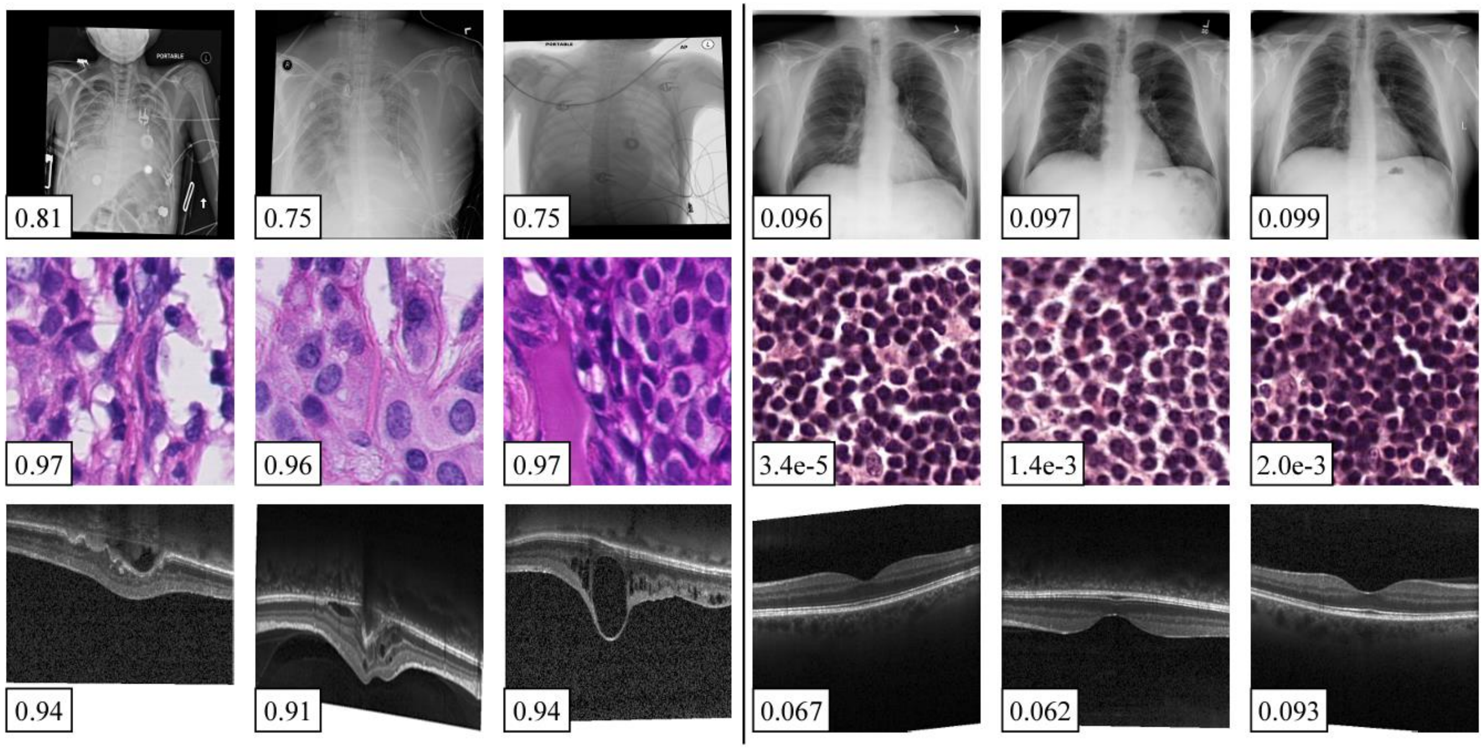
- 5개의 medical domain의 6개 dataset에 대해서 실험 진행함

Datasets	Sources	Train (all-normal)	Train (with labels)	Test	Sample size	Annotation Level
BrainMRI	BraTS2021 [1, 2, 35]	7,500	83	3,715	240*240	Segmentation mask
LiverCT	BTCV[29] + LiTs [7]	1,452	166	1,493	512*512	Segmentation mask
RESC	RESC [21]	4,297	115	1,805	512*1,024	Segmentation mask
OCT17	OCT2017 [28]	26,315	32	968	512*496	Image label
ChestXray	RSNA [51]	8,000	1,490	17,194	1,024*1,024	Image label
HIS	Camelyon16 [4]	5,088	236	2,000	256*256	Image label

Table 6. Summary of datasets from different medical modalities.

Experimental Setups

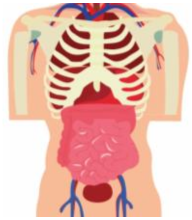
Dataset



(a) Abnormal Samples (b) Normal Samples

$$(x_i, c_i, S_i)_{i=1}^{K}$$

- c_i: image-level anomaly classification (AC) label $c_i \in \{-,+\}$.
  - '+' indicates anomalous samples
  - '-' indicates normal samples.
- S_i: pixel-level anomaly segmentation (AS) annotations $S_i \in \{-,+\}^{h\times w}$ for images of size h × w

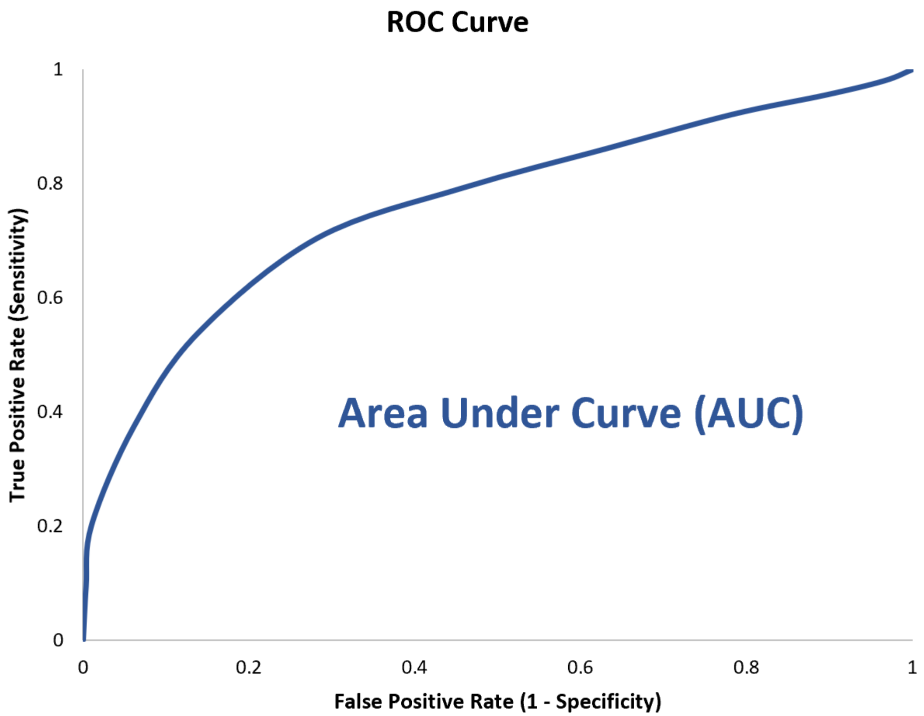


Experiments

Experimental Setups

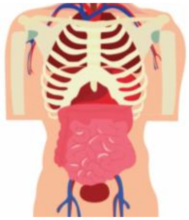
Evaluation

- *all-normal (vanilla method):* training dataset에 있는 normal image들만 모두 사용하여 학습함
- *few-normal-shot:* 적은 수의 normal image만 사용하여 학습함
- *few-shot:* 적은 수의 normal image와 적은 수의 anomaly image를 활용하여 학습함



✓ *The area under the Receiver Operating Characteristic curve metric(AUC) Metric을 통해 평가됨*

✓ Metric measures the ability of a model to distinguish between positive and negative classes with a value of 1 indicating perfect discrimination and 0.5 representing random chance. Higher AUC values signify better model performance in classification tasks.



Experiments

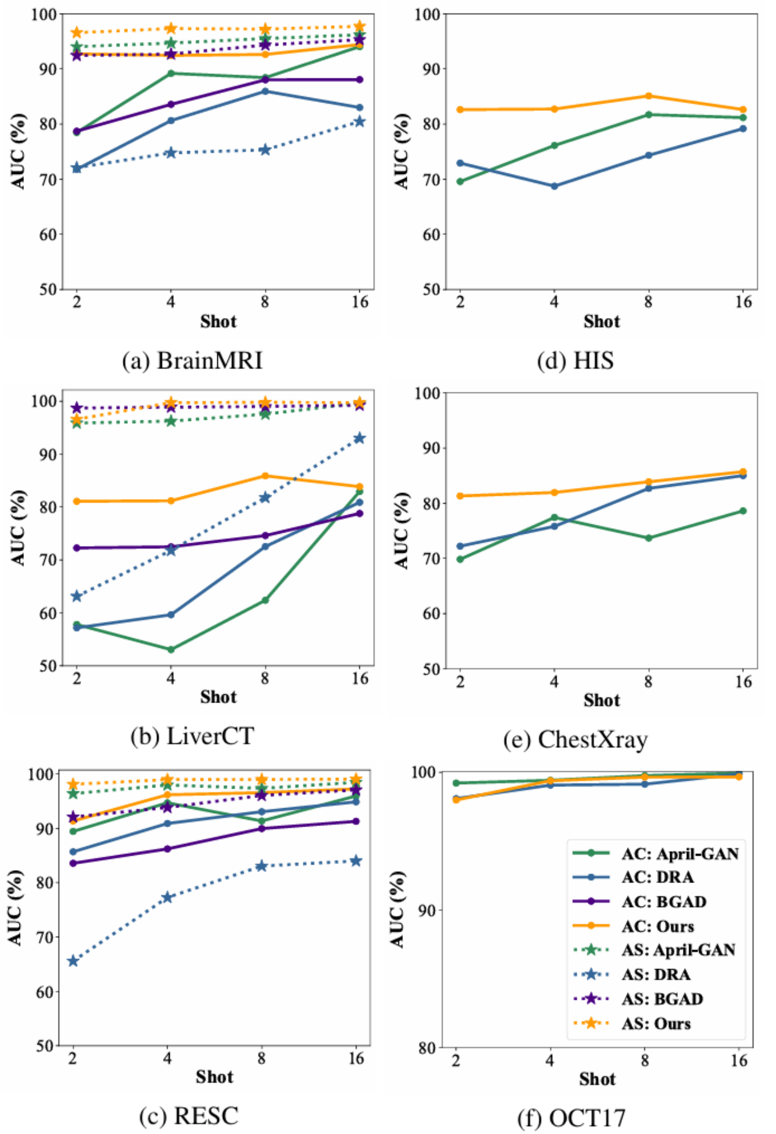
Comparison with SoTA Methods

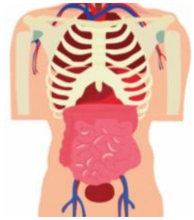
Few-shot with K = 4

- ✓ few-shot에서는 가장 좋은 성능을 보였고, 다른 setting과 비교했을 때도 우수한 성능을 보이고 있음
- ✓ full-normal-shot에서는 대용량의 normal dataset이 사용되는데, MVFA의 경우 이보다 훨씬 적은 수의 normal data만으로도 full-normal-shot setting의 결과보다 좋은 AC, AS 결과를 보임
- ✓ K 값을 달리 했을 때도 AS와 AC 모두 기존 SoTA 모델들에 비해 좋은 결과를 보이므로, MVFA가 적은 수의 abnormal image에 대해서 잘 학습하고 있다는 것을 알 수 있음

Table 1. Comparisons with state-of-the-art **few-shot** anomaly detection methods with K=4. The AUCs (in %) for anomaly classification (AC) and anomaly segmentation (AS) are reported. The best result is in bold, and the second-best result is underlined.

Setting	Method	Source	HIS	ChestXray	OCT17	BrainMRI		LiverCT		RESC	
			AC	AC	AC	AC	AS	AC	AS	AC	AS
full-normal-shot	CFlowAD [19]	WACV 2022	54.54	71.44	85.43	73.97	93.52	49.93	92.78	74.43	93.75
	RD4AD [10]	CVPR 2022	66.59	67.53	97.24	89.38	96.54	60.02	95.86	87.53	96.17
	PatchCore [41]	CVPR 2022	69.34	75.17	98.56	<u>91.55</u>	<u>96.97</u>	60.40	96.58	91.50	96.39
	MKD [44]	CVPR 2022	<u>77.74</u>	<u>81.99</u>	96.62	81.38	89.54	60.39	96.14	88.97	86.60
few-normal-shot	CLIP [25]	OpenCLIP	63.48	70.74	98.59	74.31	93.44	56.74	97.20	84.54	95.03
	MedCLIP [52]	EMNLP 2022	75.89	**84.06**	81.39	76.87	90.91	60.65	94.45	66.58	88.98
	WinCLIP [26]	CVPR 2023	67.49	70.00	97.89	66.85	94.16	67.19	96.75	88.83	96.68
few-shot	DRA [12]	CVPR 2022	68.73	75.81	99.06	80.62	74.77	59.64	71.79	90.90	77.28
	BGAD [57]	CVPR 2023	-	-	-	83.56	92.68	<u>72.48</u>	<u>98.88</u>	86.22	93.84
	APRIL-GAN [9]	arXiv 2023	76.11	77.43	**99.41**	89.18	94.67	53.05	96.24	<u>94.70</u>	<u>97.98</u>
	MVFA	Ours	**82.71**	81.95	<u>99.38</u>	**92.44**	**97.30**	**81.18**	**99.73**	**96.18**	**98.97**





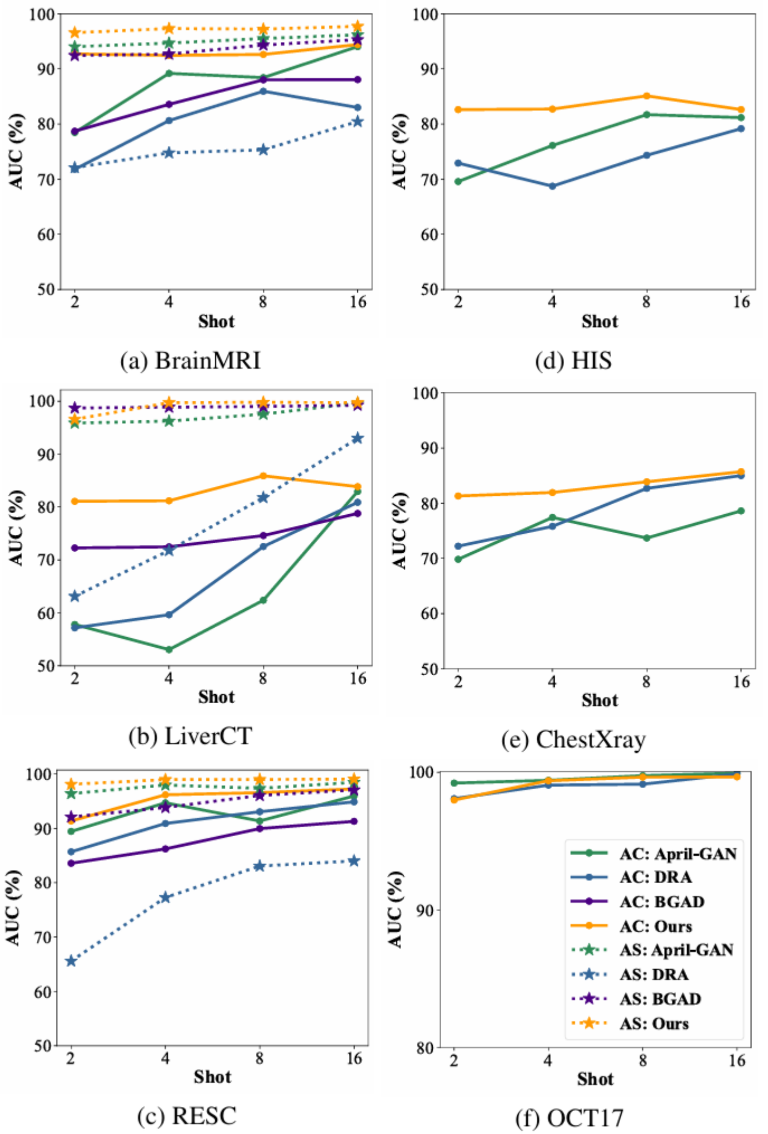
Experiments

Comparison with SoTA Methods

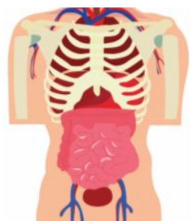
Few-shot Evaluation

Table 7. Comparisons with state-of-the-art **few-shot** anomaly detection methods with $K = 2, 4, 8, 16$. The AUCs (in %) for anomaly classification (AC) and anomaly segmentation (AS) are reported. The best result is in bold, and the second-best result is underlined.

Shot Number	Method	Source	HIS	ChestXray	OCT17	BrainMRI		LiverCT		RESC	
			AC	AC	AC	AC	AS	AC	AS	AC	AS
2-shot	DRA [12]	CVPR 2022	72.91	72.22	98.08	71.78	72.09	57.17	63.13	85.69	65.59
	BGAD [57]	CVPR 2023	-	-	-	78.70	92.42	72.27	**98.71**	83.58	92.10
	APRIL-GAN [9]	arXiv 2023	69.57	69.84	**99.21**	78.45	94.02	57.80	95.87	89.44	96.39
	MFA	ours	**82.61**	**81.32**	97.98	**92.72**	**96.55**	**81.08**	96.57	**91.36**	**98.11**
4-shot	DRA [12]	CVPR 2022	68.73	75.81	99.06	80.62	74.77	59.64	71.79	90.90	77.28
	BGAD [57]	CVPR 2023	-	-	-	83.56	92.68	72.48	98.88	86.22	93.84
	APRIL-GAN [9]	arXiv 2023	76.11	77.43	**99.41**	89.18	94.67	53.05	96.24	94.70	97.98
	MFA	ours	**82.71**	**81.95**	99.38	**92.44**	**97.30**	**81.18**	**99.73**	**96.18**	**98.97**
8-shot	DRA [12]	CVPR 2022	74.33	82.70	99.13	85.94	75.32	72.53	81.78	93.06	83.07
	BGAD [57]	CVPR 2023	-	-	-	88.01	94.32	74.60	99.00	89.96	96.06
	APRIL-GAN [9]	arXiv 2023	81.70	73.69	**99.75**	88.41	95.50	62.38	97.56	91.36	97.36
	MFA	ours	**85.10**	**83.89**	99.64	**92.61**	**97.21**	**85.90**	**99.79**	**96.57**	**99.00**
16-shot	DRA [12]	CVPR 2022	79.16	85.01	99.87	82.99	80.45	80.89	93.00	94.88	84.01
	BGAD [57]	CVPR 2023	-	-	-	88.05	95.29	78.79	99.25	91.29	97.07
	APRIL-GAN [9]	arXiv 2023	81.16	78.62	**99.93**	94.03	96.17	82.94	99.64	95.96	98.47
	MFA	ours	**82.62**	**85.72**	99.66	**94.40**	**97.70**	**83.85**	**99.73**	**97.25**	**99.07**



(a) BrainMRI
(d) HIS
(b) LiverCT
(e) ChestXray
(c) RESC
(f) OCT17



Experiments

Comparison with SoTA Methods

Zero-shot Evaluation

- ✓ Zero-shot 환경에서의 성능평가는 *leave-one-out 설정*을 기반으로 진행됨
  - ✓ **Leave-one-out:** 하나의 데이터셋을 평가할 때 해당 데이터셋을 제외한 다른 모든 데이터셋으로 모델을 학습시키고, 배제된 데이터셋으로 테스트하는 방법. 이를 통해 학습 시 보지 못한 modality나 anatomical region에 대한 결과를 확인할 수 있으며, 이는 모델의 일반화 능력을 평가하는 데 유용함
- ✓ 실험 결과, MVFA가 zero-shot AC와 AS에서 모두 좋은 성능을 보이고 있다는 것을 알 수 있음

Table 2. Comparisons with state-of-the-art **zero-shot** anomaly detection methods with in-/out-domain evaluation. The AUCs (in %) for AC and AS are reported.

Datasets	WinCLIP [26]	April-GAN [9]	MVFA (ours)
HIS	69.85 / -	72.36 / -	**77.90** / -
ChestXray	70.86 / -	57.49 / -	**71.11** / -
OCT17	46.64 / -	92.61 / -	**95.40** / -
BrainMRI	66.49 / 85.99	76.43 / **91.79**	**78.63** / 90.27
LiverCT	64.20 / 96.20	70.57 / 97.05	**76.24** / **97.85**
RESC	42.51 / 80.56	75.67 / 85.23	**83.31** / **92.05**

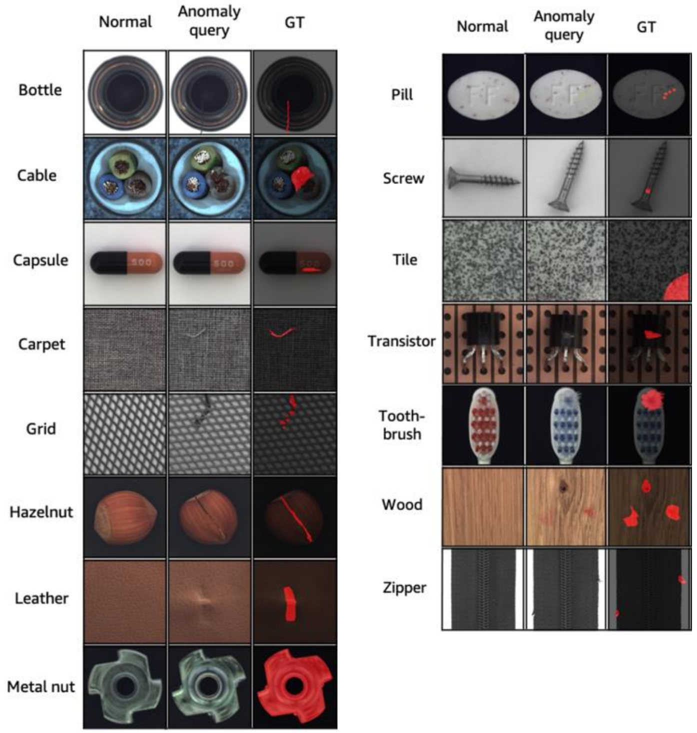
Comparison with SoTA Methods

In-domain Evaluation

- ✓ 또한 MVTec이라는 산업용 Anomaly Detection 데이터셋에 대해서 **in-domain evaluation**을 실행함
- ✓ MVFA는 in-domain evaluation을 위한 모델이 아님에도 불구하고, K=4라는 few-shot 환경에서 좋은 in-domain 성능을 보였음
- ✓ 이를 통해서 MVFA 모델 자체가 다양한 도메인에 적용될 수 있는 generalized model이라는 것을 증명함

Table 3. Comparisons with state-of-the-art **few-shot** anomaly detection methods with K=4 for in-/out-domain evaluation. The AUCs (in %) for AC/AS are reported.

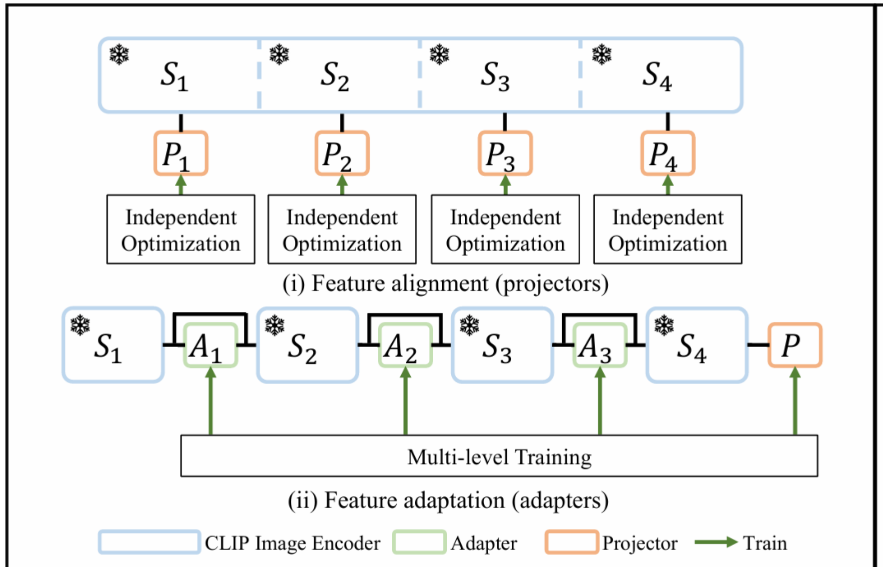
AC/AS (avg AUC%)	WinCLIP [26]	April-GAN [9]	MVFA
in-domain (MVTec)	95.16/96.27	92.77/95.89	**96.19/96.32**
out-domain (medical)	76.38/95.86	81.65/96.30	**88.97/98.67**



Experiments

Ablation Studies

Feature Adaptation vs. Feature Alignment



(a) Ablation architectures of Table 4 in the main paper

- ✓ **Feature Alignment:** Feature Adapter 대신에 Projector들만 사용함
- ✓ **Feature Adaptation:** Multi-level Feature Adapter를 사용함

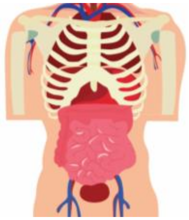
zero-shot setting에 대해서 AC와 AS 성능을 도출했을 때,

Adapter를 사용했을 때 AC와 AS 모두에서 더 높은 성능을 보였는데,

이는 **multi-level feature가 model의 generalization 성능을 촉진**했다는 것을 반증함

Table 4. Ablation studies compared with feature alignment with multi-level projectors and feature adaptation with multi-level adapters under the zero-shot setting. The AUCs (in %) for AC and AS are reported.

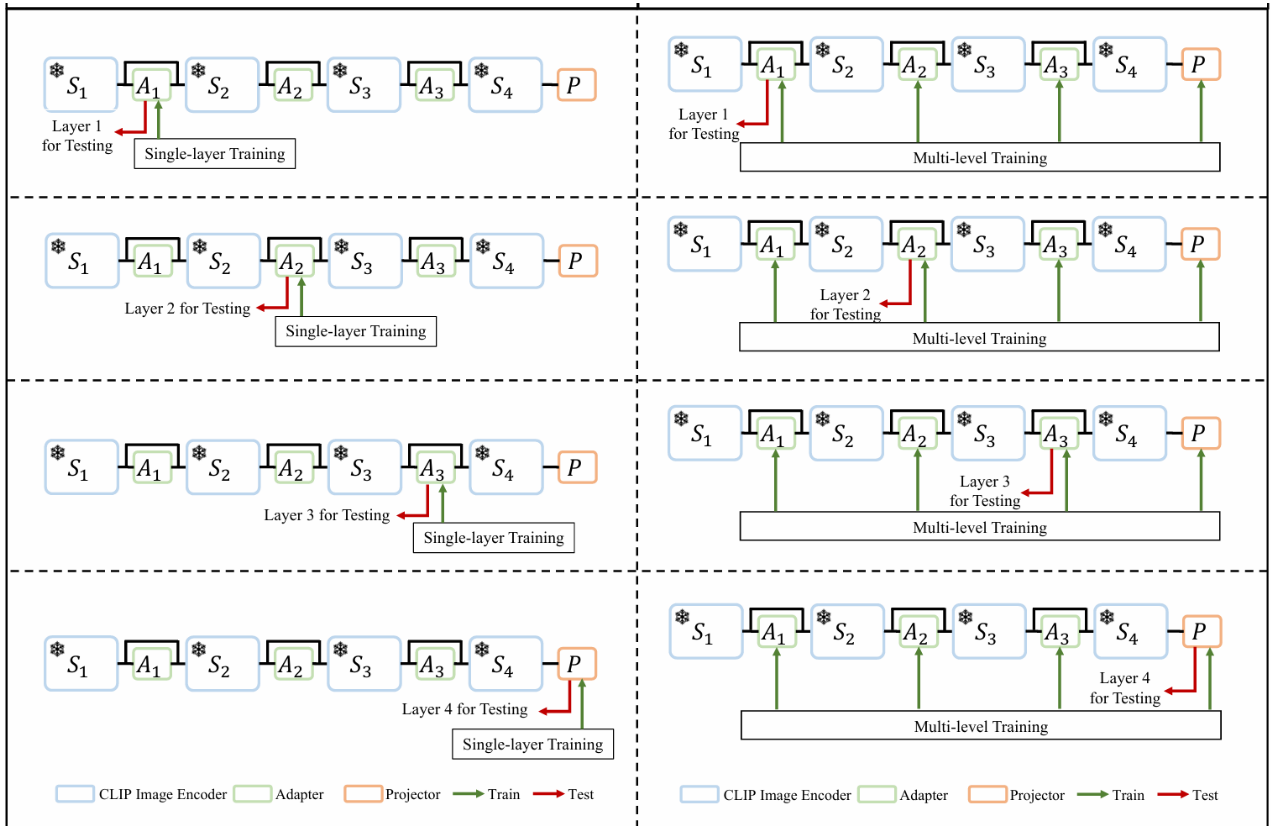
Method	HIS	ChestXray	OCT17	BrainMRI		LiverCT		RESC	
	AC	AC	AC	AC	AS	AC	AS	AC	AS
feature alignment (projectors)	66.32	58.06	49.85	76.66	89.39	75.85	97.64	74.44	89.17
feature adaptation (adapters)	**77.90**	**71.11**	**95.40**	**78.63**	**90.27**	**76.24**	**97.85**	**83.31**	**92.05**



Experiments

Ablation Studies

Feature Ensemble in Multi-level



(i) Individual layers for testing under single-layer training

(ii) Individual layers for testing under multi-level training

(c) Ablation architectures of Table 5 in the main paper

- ✓ *single-layer training*: adaptor에서 하나의 layer만 사용하여 optimize되는 것
- ✓ *multi-level training*: adaptor의 다양한 layer에서 나온 feature들을 모두 사용하여 optimize하는 것

Table 5. Ablation studies of multi-level feature ensemble under *single-level training/multi-level training* with the same model architecture. The average AUCs (in %) of all six datasets for AC and AS under the few-shot setting (K=4) are reported. Results for each dataset are included in Appendix C.

	Layer 1	Layer 2	Layer 3	Layer 4	All
AC	80.96/83.39	88.83/88.84	81.25/84.32	83.33/84.62	**88.97**
AS	97.70/97.98	98.58/98.62	96.03/98.54	97.19/97.44	**98.67**

Experiments

Ablation Studies

Feature Ensemble in Multi-level

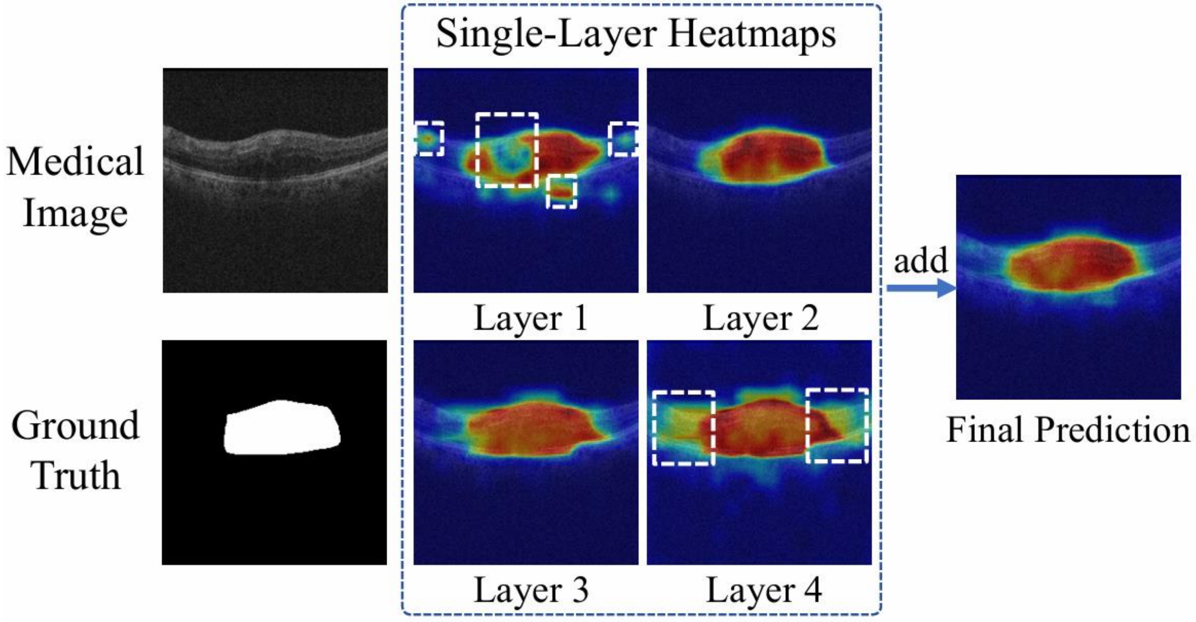


Figure 4. Considering features across multiple levels significantly enhances segmentation performance. The white dashed boxes demarcate regions that have been missed or erroneously segmented. More cases are included in Appendix D.

- ✓ *single-layer training*: adaptor에서 하나의 layer만 사용하여 optimize되는 것
- ✓ *multi-level training*: adaptor의 다양한 layer에서 나온 feature들을 모두 사용하여 optimize하는 것

Table 5. Ablation studies of multi-level feature ensemble under *single-level training/multi-level training* with the same model architecture. The average AUCs (in %) of all six datasets for AC and AS under the few-shot setting (K=4) are reported. Results for each dataset are included in Appendix C.

	Layer 1	Layer 2	Layer 3	Layer 4	All
AC	80.96/83.39	88.83/88.84	81.25/84.32	83.33/84.62	**88.97**
AS	97.70/97.98	98.58/98.62	96.03/98.54	97.19/97.44	**98.67**

Experiments

Visualization Analysis

Anomaly Segmentation (AS)

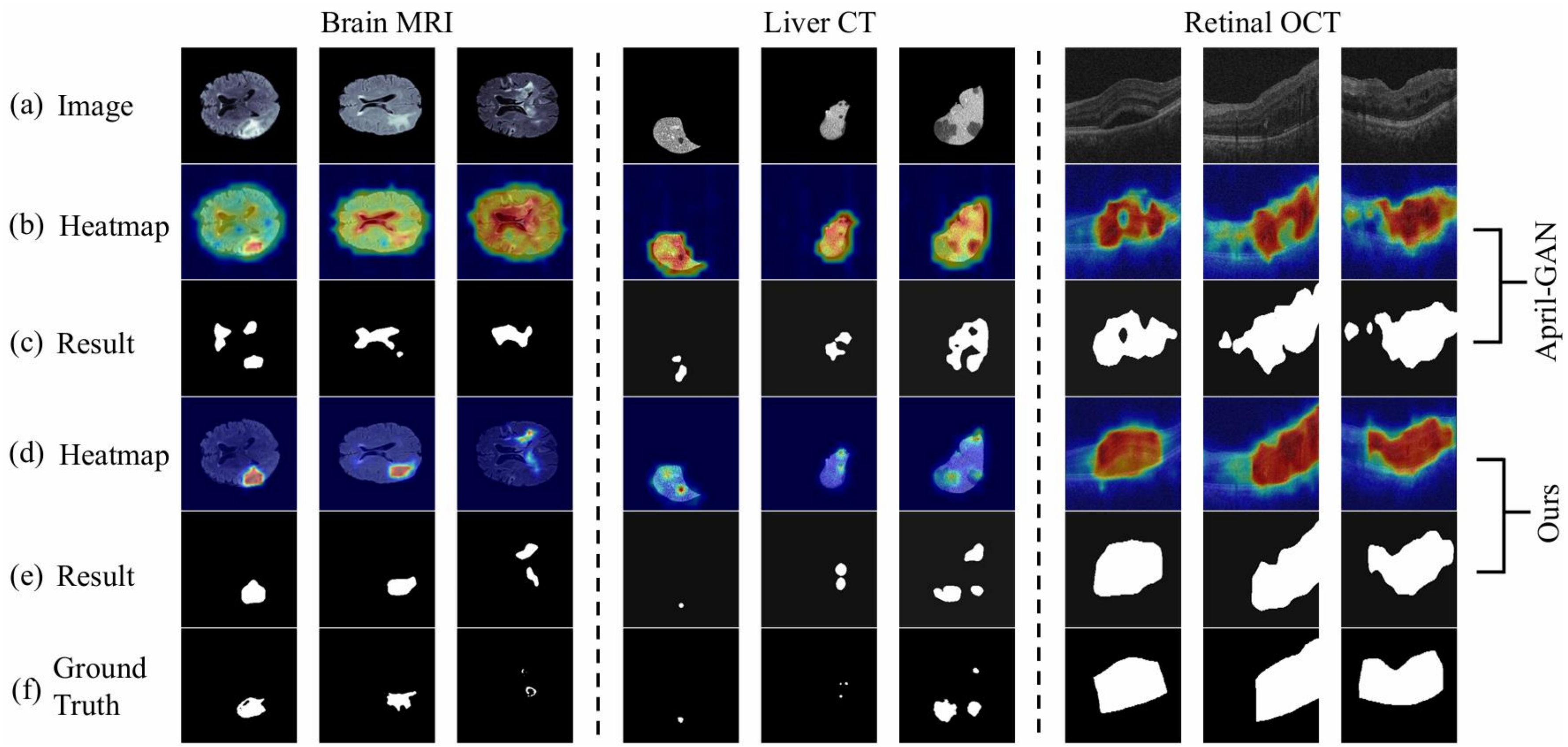
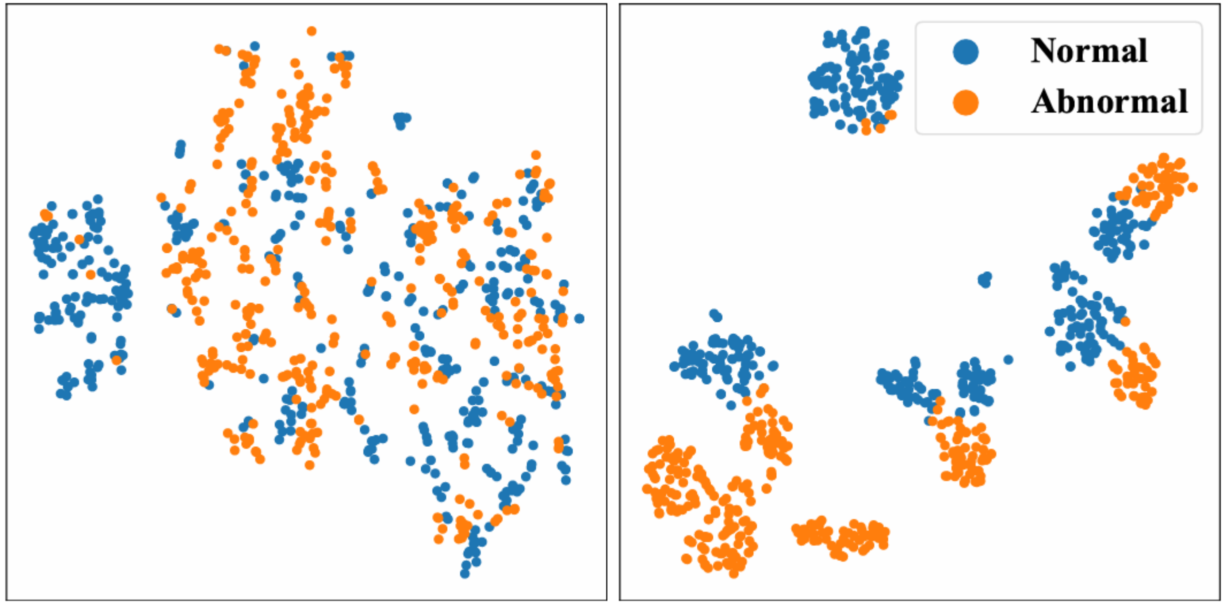


Figure 5. Visualization of AS for MVFA on brain MRI, liver CT, and retinal OCT, compared with state-of-the-art method April-GAN. Results from (e) show better performance than results from (c), showing the effectiveness of the proposed multi-level feature adaptation.

Experiments

Visualization Analysis

t-SNE Results



(a) Without Adapter (b) With Adapter

Figure 6. Visualization, using t-SNE, of the features learned from the RESC test set, using (a) pretrained visual encoder, and (b) multi-level feature adapters. The same t-SNE optimization iterations are used in each case. Results show that features extracted by adapters are separated between normal and abnormal samples.

- ✓ t-SNE를 비교해 보면 adapter의 유무에 따라서 normal과 abnormal에 대한 feature alignment가 어떻게 달라지는지 확인할 수 있음
- ✓ Adapter가 있을 때 normal과 abnormal에 대한 decision boundary가 잘 형성되어 있음
- ✓ 반면 adapter가 없을 때는 feature가 제대로 align되지 않았다는 것을 시각적으로 확인할 수 있었다.

Conclusion

01

CLIP 모델을 Medical Image Anomaly Detection에 맞게 재구성 하기 위해서 **Multi-level Adaptation and Comparison Framework**를 도입함. 이는 *natural image로 pretrain된 VLM 모델의 Vision Encoder에 있는 여러 개의 residual adapter들을 통합하여 multi-level feature들을* 활용할 수 있도록 함

02

다양한 level의 feature들을 활용하기 위해서 Multi-level, Pixel-wise Visual-Language Feature들을 관장할 수 있는 Loss Function을 활용함. 이는 모델의 초점을 자연 이미지의 객체 의미에서 의료 이미지의 Anomaly Detection로 guide하는 역할을 하게 됨.

03

본 연구는 일반화 가능한 의료 이상 탐지 모델을 개발하여 다양한 의료 데이터 유형에서 개선된 일반화를 보여줌. 특히, 제안된 방법은 다양한 modality(brain MRI, liver CT, retinal OCT, chest X-ray, and digital histopathology)의 zero-/few-shot 시나리오에서 현재의 최신 모델들을 능가함.

Q & A

Thank you for listening